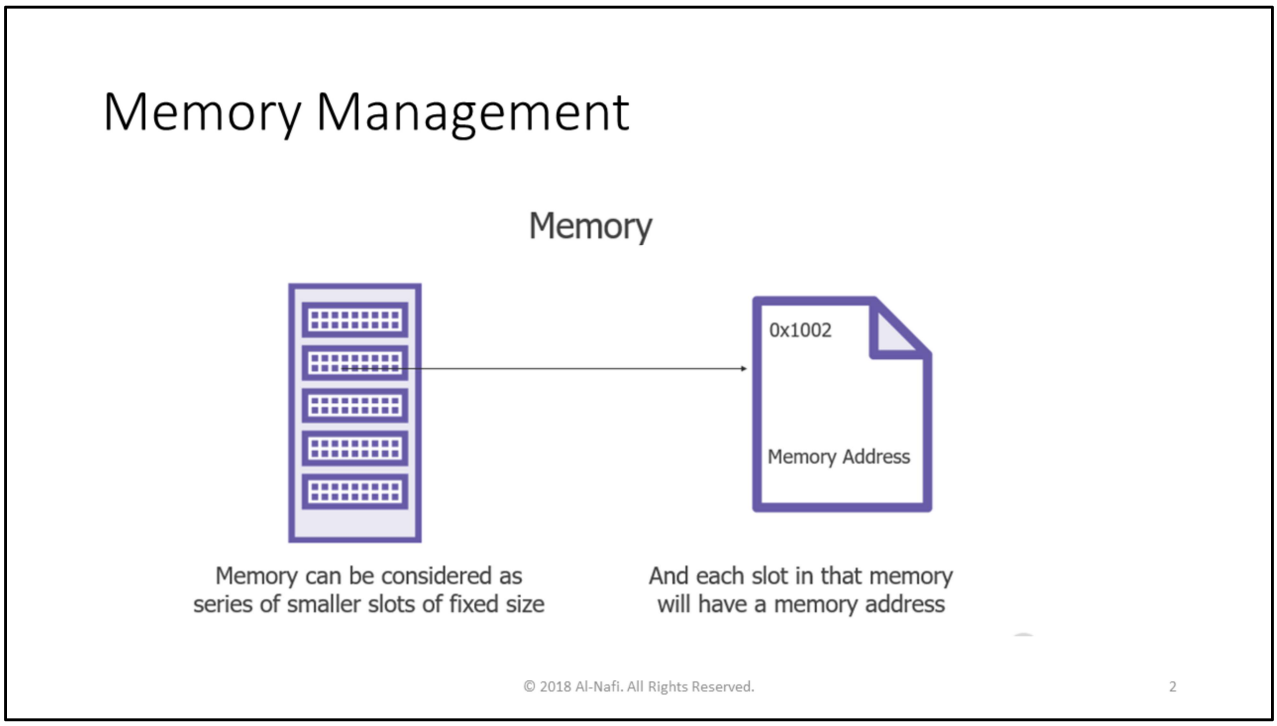
Access Control



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Modern systems include some form of access control. Even kiosk or general user type systems internally implement a system of permissions and rules for accessing processes, memory, applications, and operating system functions even if those controls are transparent to the end user. Access controls are typically enforced by a kernel level module known as the security monitor or reference monitor.

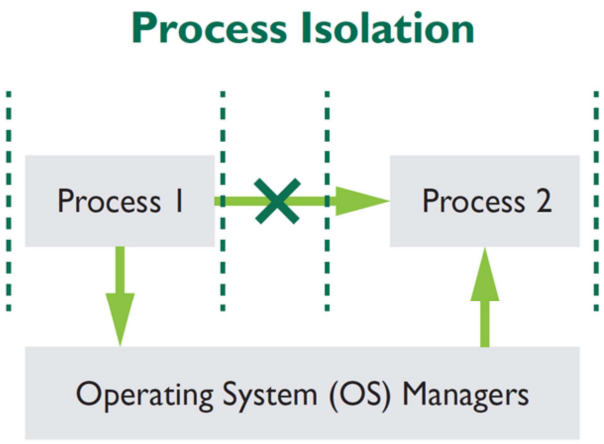
Access control mechanisms are typically supported by the file system that often stores security attributes with files and enables fine-grained access control in storage objects.



From a security perspective, memory and storage are the most important resources in any computing system. Ideally, it would be possible to easily separate memory used by subjects (such as running processes and threads) from objects (such as data in storage). Buffer overflows are a common type of attack that attempts to write executable code into memory locations where it may be inadvertently executed.

Modern operating systems utilize a variety of techniques to limit the exposure of the memory space to a potential attacker. Direct access to the system memory is limited to user-space programs or allocated randomized blocks of memory space to limit the utility of a crafted memory attack running within a program or piece of code. Additionally, memory space for user programs may be monitored by the operating system to ensure it is utilizing memory properly and that executable code is only located in authorized memory blocks. An example is Data Execution Prevention (DEP) technology in Windows that will close a program or code that is mismanaging memory or attempting to execute code from unauthorized locations.

Process Isolation



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Processors and their supporting chipsets provide one of the first layers of defense in any computing system. In addition to providing specialized processors for security functions (such as cryptographic coprocessors), processors also have states that can be used to distinguish between more or less privileged instructions.

Most processors support at least two states:

- A supervisor state
- A problem state

In supervisor state (also known as kernel mode), the processor is operating at the highest privilege level on the system, and this allows the process running in supervisor state to access any system resource (data and hardware) and execute both privileged and non-privileged instructions.

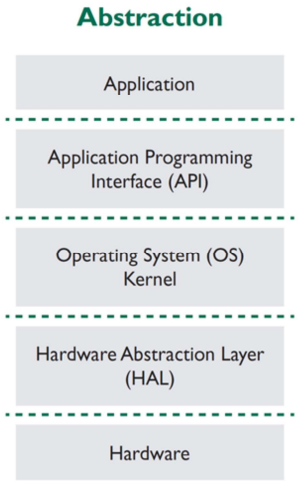
In problem state (also known as user mode), the processor limits the access to system data and hardware granted to the running process. A malicious process running in supervisor state has very few restrictions placed upon it and can be used to cause a lot of damage. Ideally, access to supervisor state is limited only to core

OS functions that are abstracted from end-user interaction through other controls, but this is not always the case.

Process isolation can also be used to prevent individual processes from interacting with each other. This can be done by providing distinct address spaces for each process, and preventing other processes from accessing that area of memory, and assigning access permissions to files or other resources to each process.

Naming distinctions are also used to distinguish between different processes. Virtual mapping is also used to assign randomly chosen areas of actual memory to a process to prevent other processes from finding those locations easily. Encapsulation of processes as objects can also be used to isolate them, since an object includes the functions for operating on it, the details of how it is implemented can be hidden. The system can also ensure that shared resources are managed to ensure that processes are not allowed to access shared resources in the same time slots.

Abstraction Layers



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Abstraction involves the removal of characteristics from an entity to easily represent its essential properties. Abstraction negates the need for users to know the particulars of how an object functions. They only need to be familiar with the correct syntax for using an object and the nature of the information that will be presented as a result. Since a separate subject controls the access to the object, the ability to manipulate the object outside of the defined rules is limited.